



Name _____ Period ____ Test Day _____

Unit 12 - Solutions and Acids/Base Chemistry Essential Knowledge and Guided Notes

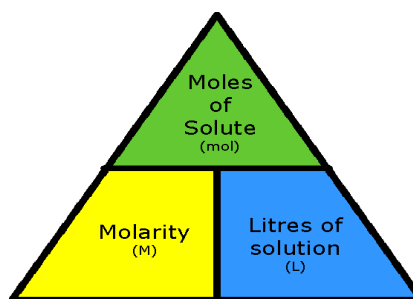
Textbook: CK Chemistry Chapter 16 - Solutions & Chapter 21 - Acids & Bases
<https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/>

Helpful Videos and Practice: Posted on CANVAS

U12 Part 1 - Solutions Essential Knowledge

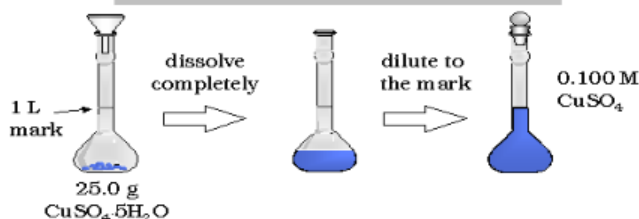
1. Solutions can be a variety of solute/solvent combinations: gas/gas, gas/liquid, liquid/liquid, solid/liquid, gas/solid, liquid/solid, or solid/solid. The solvent is always the part that is in larger amounts. A common solvent is water.
2. Solubility curves show how much of a solute can be dissolved in a solvent before saturation is reached. Saturated Solutions contain the maximum amount of solute at a given temperature that can be dissolved. The saturation of a solution is dependent on the amount of solute present in the solution.
3. Temperature can have a great effect on solubility. Solubility graphs relate solubility to temperature. For most solid solutes, as temperature increases, the solubility increases. For most gas solutes, solubility decreases as temperature increases.
4. The unit of Concentration used in chemistry is

$$\text{Molarity (M)} = \frac{\text{Moles of solute (mol)}}{\text{Liters of Solutions (L)}}$$



$$\text{molarity (M)} = \frac{\text{moles of solute}}{\text{liters of solution}}$$

5. Process of making a solution using a Volumetric flask:

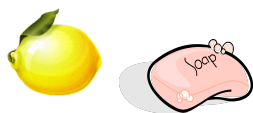


6. When solutions are diluted, the moles of solute present initially remain. You can determine the molar concentration of a diluted solution using the relationship:

$$M_{\text{initial}} V_{\text{initial}} = M_{\text{final}} V_{\text{final}}$$

$$M_1 V_1 = M_2 V_2$$

7. When diluting Acids, add the acid to the water to prevent an exothermic splash in the face of concentrated acid. That's because acid dissolving into water produces so much heat that the water can turn to steam and splatter the solution. If acid is added to water, the liquid that splatters will be mostly water. If water is added to acid, the liquid that splatters will be mostly acid. Also, water has a high ability to absorb heat. So adding acid to water will be cooled by the extra water. If you want to be crazy, then do the opposite! "Do what you oughta, add acid to water!"



U12 Part 2 - Acid/Base Chemistry Essential Knowledge

Definition and Properties of Acids and Bases

	Acid	Bases
Arrhenius Model	An acid is any substance that adds hydrogen ion (H^+) to solution.	A base is any substance that adds hydroxide (OH^-) to solution.
Bronsted-Lowry Model	An acid is a proton (H^+) donor.	A base is a proton (H^+) acceptor.
Example	$HCl + H_2O \rightarrow H_3O^+ + Cl^-$	$NaOH \rightarrow Na^+ + OH^-$ $NH_3 + H_2O \rightarrow NH_4^+ + OH^-$
Properties	Taste Sour Low pH Turn Litmus Paper Red	Taste Bitter, Feels Slippery High pH Turns Litmus Paper Blue

Acidic Solution: a solution containing more hydrogen ions [H^+] than hydroxide ions [OH^-].

Basic Solution: a solution containing more hydroxide ions [OH^-] than hydrogen ions [H^+].

pH and pOH

Relationship among [H^+], [OH^-], and pH			
	[H^+] (mol/L)	[OH^-] (mol/L)	pH Aqueous system
Increasing acidity ↑	1×10^0	1×10^{-14}	0.0 ← 1M HCl
	1×10^{-1}	1×10^{-13}	1.0 ← 0.1M HCl
	1×10^{-2}	1×10^{-12}	2.0 ← Gastric juice
	1×10^{-3}	1×10^{-11}	3.0 ← Lemon juice
	1×10^{-4}	1×10^{-10}	4.0 ← Tomato juice
	1×10^{-5}	1×10^{-9}	5.0 ← Black coffee
Neutral	1×10^{-6}	1×10^{-8}	6.0 ← Milk
	1×10^{-7}	1×10^{-7}	7.0 ← Pure water
	1×10^{-8}	1×10^{-6}	8.0 ← Blood
	1×10^{-9}	1×10^{-5}	9.0 ← Sodium bicarbonate, sea water
	1×10^{-10}	1×10^{-4}	10.0 ← Milk of magnesia
Increasing basicity ↓	1×10^{-11}	1×10^{-3}	11.0 ← Household ammonia
	1×10^{-12}	1×10^{-2}	12.0 ← Washing soda
	1×10^{-13}	1×10^{-1}	13.0 ← 0.1M NaOH
	1×10^{-14}	1×10^0	14.0 ← 1M NaOH

What is pH?

A number scale used to indicate the acidity and strength of an acid or base

$$pH = -\log [H^+]$$

$$[H^+] = 10^{-pH}$$

Acidic Solution: $pH < 7.0$ [H^+] more than $1 \times 10^{-7} M$

Neutral Solution: $pH = 7.0$ [H^+] equal to $1 \times 10^{-7} M$

Basic Solution: $pH > 7.0$ [H^+] less than $1 \times 10^{-7} M$

What is pOH?

A number scale used to indicate the alkalinity (basicity) and strength of an acid or base

$$pOH = -\log [OH^-]$$

$$[OH^-] = 10^{-pOH}$$

How are pH and pOH related?

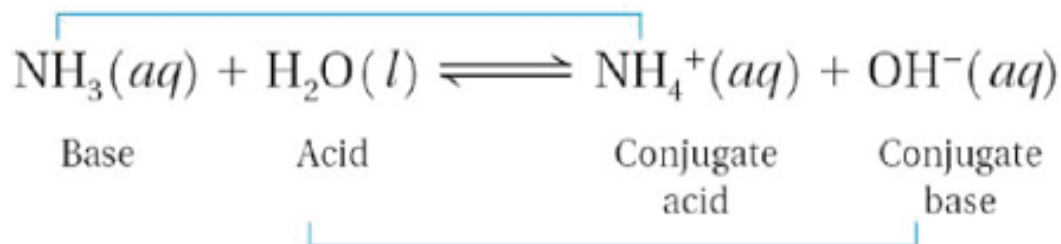
$$pH + pOH = 14$$

$$[OH^-] \times [H^+] = 1.0 \times 10^{-14}$$

Definition of Conjugate Acids and Conjugate Bases

A conjugate acid is the particle formed when a base gains a hydrogen ion.

A conjugate base is the particle that remains when an acid has donated a hydrogen ion.



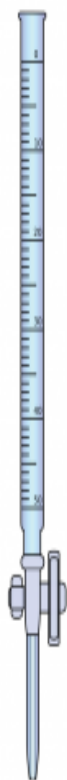
Neutralization Reactions result from the reaction of an acid with a base to form a salt and water. These reactions are double replacement reactions. Salts are ionic compounds.

Ex: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{HOH}$ and $\text{HNO}_3 + \text{KOH} \rightarrow \text{KNO}_3 + \text{HOH}$
Neutralization occurs when $[\text{H}^+] = [\text{OH}^-]$

For neutralization of strong acids with strong bases, the neutralization pH is 7.

Titration is a laboratory technique in which you determine the molar concentration of an unknown acid or base using a standard solution of a known acid or base in combination with an indicator. An indicator is a substance that changes color based on the pH of the solution letting you know when you have reached a neutralization point which is called the "endpoint" of the titration. For titration, a buret is used to dispense precise amounts of solution of known concentration to determine the concentration of another solution. The endpoint of a titration can be seen when the pH indicator changes color.

For solutions with an equal number of moles of OH^{1-} and H^{1+} use this to calculate the unknown: $M_{\text{acid}}V_{\text{acid}} = M_{\text{base}}V_{\text{base}}$
Otherwise, use stoichiometry to determine the mole ratio of acid to base to determine the concentration of the unknown.



U12 Lesson #1 Solubility Curves

1. Copy the Definitions

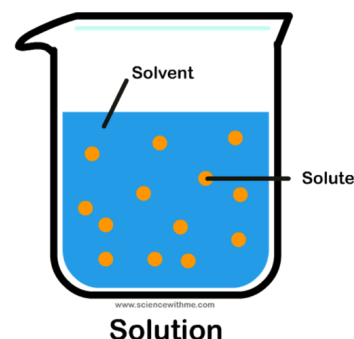
Solution:

Solvent:

Solute:

Unsaturated Solution:

Saturated Solution:

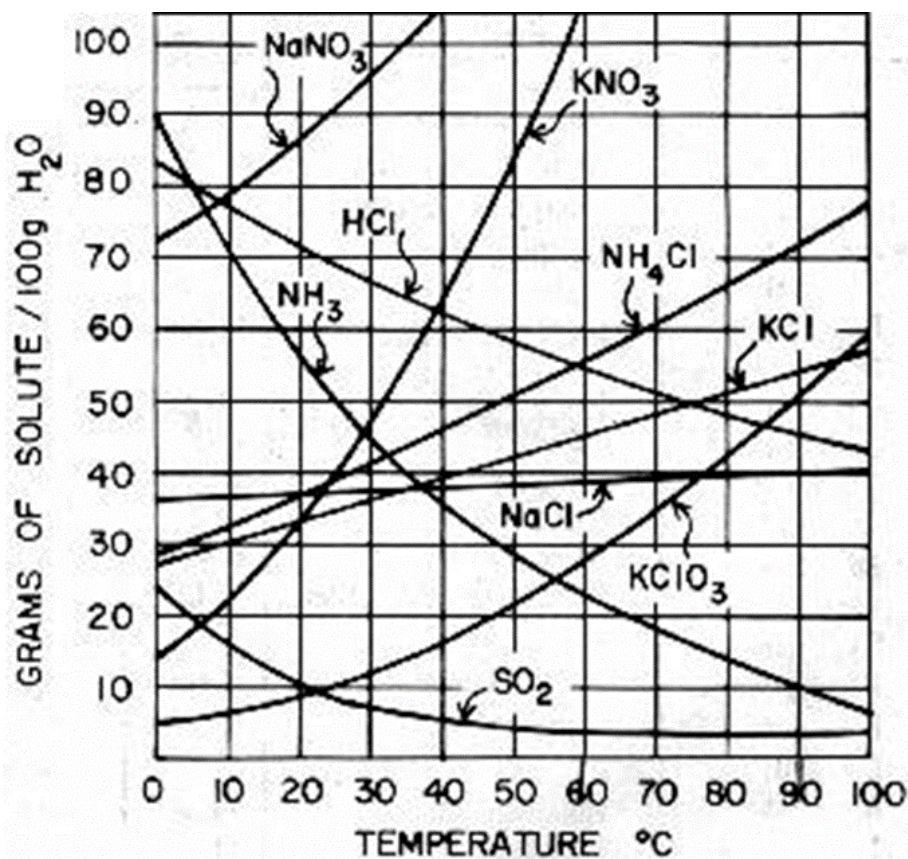


Supersaturation is an unnatural state. Generally, the excess amount of solute will not dissolve and remain on the bottom of the container

Here is a Solubility Curve: It shows how much solute it takes to saturate a certain amount of solvent at specific temperatures. **Solids** get more soluble as the temperature increases. Notice the lines go up with temperature (Ionic compounds are solids! Formulas with Metals and Non-Metals) **Gases** get less soluble as the temperature increases. Notice the lines go down. (NH_3 and HCl , SO_2 are gases.)

Questions:

1. Independent Variable? _____
2. Dependent Variable? _____
3. Look at the y-axis, what is the unit of solubility? _____
4. At 70°C , how much NH_4Cl can be dissolved in 100 g of H_2O ? _____
5. At 70°C , how much NH_4Cl can be dissolved in 400 g of H_2O ? _____
6. At 70°C , how much NH_4Cl can be dissolved in 175 g of H_2O ? _____

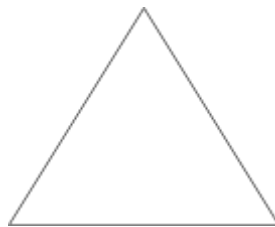


U12 Lesson #2 Molarity and Dilution

To express the concentration of solutions in Chemistry use _____

For solutions with a solid solute and a liquid solvent:

Molarity (M) = $\frac{\text{moles of solute (mol)}}{\text{liters of solution (L)}}$



M =

mol =

L =

Examples:

What is the molarity (M) if 4.0 L solution contains 2.0 moles of CuCl_2 ? _____

Work:

How many moles are needed to make 3.0 L of a 0.50 M solution of NaBr? _____

Work:

How many liters is a 5.0 M solution containing 10.0 moles of KCl? _____

Work:

How many grams of KNO_3 (101.11 g/mol) to prepare 4.00 L of a 0.500 M solution? _____

Work:

Molarity can be changed by _____

As a solution is diluted, the molarity _____

Stock solutions are _____ and _____

Formula for Dilutions. Moles (Initial) = Moles (Final)

$$M_1V_1 = M_2V_2$$

Examples:

A scientist needs 500. ml of 1.0 M HCl (hydrochloric acid). The stock solution on the shelf is 12 M HCl, how much of this stock solution will he need to dilute to 500. ml in order to get the solution he needs.

Joe Chemistry Student has 100. ml of 6.0 M HCl solution. What will the new concentration be if he dilutes it to 1.000 L?

Safety: To dilute an acid, add the a _____ to the _____.

"Do what you oughta add acid to water!!"

U12 Lesson #3 Acids & Bases, pH, pOH Calculations

Video: (If you missed class, you can watch it on CANVAS)

Properties of Acids and Bases

1. Acids taste _____
2. Bases taste _____
3. Acids feel _____
4. Bases feel _____
5. Acids and bases are _____ which means they conduct electricity in water.
6. Acids react with _____ to make H_2 gas, but bases do not.
7. Litmus indicator paper turns _____ in bases.
8. Litmus indicator paper turns _____ in acids.

Arrhenius Definition of Acids and Bases

9. _____ is when acids react with bases to form water and salt.
10. Arrhenius Acids generate _____ ions or protons when dissolved in water.
11. Arrhenius Bases generate _____ ions when dissolved in water.
12. Acid + Base _____ + _____
13. $HCl + NaOH$ _____ + _____

Bronsted-Lowry Acids and Bases

14. Bronsted Lowry Acids are substances that _____ H^+ ions or protons.
15. Bronsted Lowry Bases are substances that _____ H^+ ions or protons.
16. Acids _____ concentration of H^+ ions.
17. Bases _____ concentration of H^+ ions

Strong and Weak Acids and Bases

18. A strong acid like HCl dissociates or readily transfers _____% of its hydrogen ions to water
19. A weak acid like vinegar only dissociates or transfers _____% of its hydrogen ions to water.

Conjugate Acid Base Pairs.

20. Conjugate acid base pairs differ by a _____ ion or proton
21. A strong acid has a _____ conjugate base.
22. A strong base has _____ conjugate acid

pH Scale

23. H_2O can act as an _____ or a _____.
24. The pH scale is based on the concentration of _____ ions.
25. Water has an H^+ concentration of _____ moles/ liter
26. Acids have a pH less than _____
27. Bases have a pH greater than _____.

Slides:

1. Characteristics of Acids and Bases:

	Acids	Bases
Taste...		
Feel...		
pH compared to 7		
Conduct electricity?		
Turn litmus paper...		

2. Examples:

Examples of Strong Acids with Formulas	Examples of Strong Bases with Formulas
Hydrochloric Acid -	Sodium Hydroxide -
Sulfuric Acid -	Potassium Hydroxide -
Nitric Acid -	Magnesium Hydroxide -
Ion present is _____ Name: Sometimes written as _____ Name:	Ion present is _____ Name:

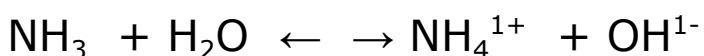
3. Definitions of Acids and Bases:

Arrhenius Acids	Bronsted-Lowry Acids
Definition:	Definition:
Example:	Examples:
Arrhenius Bases	Bronsted-Lowry Bases
Definition:	Definition:
Example:	Examples:

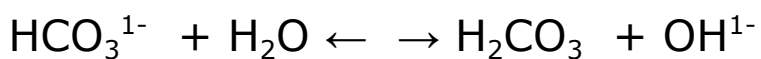
4. Conjugate Acid/Base Pairs (differ by a proton or H^{1+} ion)

Equation	Acid	Conjugate Base
$HCl \longleftrightarrow H^{1+} + Cl^{1-}$	HCl	Cl^{1-}
$HF \longleftrightarrow H^{1+} + F^{1-}$		
$H_2SO_4 \longleftrightarrow H^{1+} + HSO_4^{1-}$		
$NH_4^{1+} \longleftrightarrow H^{1+} + NH_3$		

5. Two Weak Bases to know: _____ and _____
Examine the reactions in water and label conjugate acid-base pairs:



Base Acid _____ _____ (Note: adds OH^{1-} to solution)



Base Acid _____ _____ (Note: adds OH^{1-} to solution)

6. What does pH measure? _____

7. What are the typical pH values for Acids _____ Neutral _____ Bases _____

8. **pH = $-\log [H^{1+}]$** What do brackets $[H^{1+}]$ mean? _____

$[H^{1+}]$ M (exponent form)	$[H^{1+}]$ M (decimal form)	pH	Acid or Base
1×10^{-1} M	0.1 M		Very Acidic
1×10^{-2} M	0.01 M		Acidic
1×10^{-5} M	0.00001 M		Slightly Acidic
1×10^{-7} M	0.0000001 M		Neutral
1×10^{-11} M	0.00000000001 M		Basic
1×10^{-14} M	0.00000000000001 M		Very Basic

a. Calculate the pH of 0.010 M HCl

b. Calculate the pH of 0.030 M HBr

9. What do changes in pH units mean?

Increase 1 pH Unit Less Acidic	concentration of H^{1+} decreases 10^1 or 10 times ($1 \rightarrow 2$ means 10^{-1} to 10^{-2} M)
Decrease 1 pH Unit More Acidic	concentration of H^{1+} increases 10^1 or 10 times ($2 \rightarrow 1$ means $10^{-2} \rightarrow 10^{-1}$ M)
Increase 2 pH Units	concentration of H^{1+}
Decrease 2 pH Units	concentration of H^{1+}
Increase 3 pH Units	concentration of H^{1+}
Decrease 3 pH Units	concentration of H^{1+}

10. What does pOH measure? _____

11. **pOH = - log [OH¹⁻]**

12. Relative amounts of [H⁺] and [OH⁻]

- In an acidic solution [H⁺]_____ [OH⁻]
- In a basic solution [H⁺]_____ [OH⁻]
- In a neutral solution [H⁺]_____ [OH⁻] = 1×10^{-7} M (at 25° C)

13. Relationships between [H⁺] and [OH⁻] at 25°C

- [H⁺] x [OH⁻] = _____
- pH + pOH = _____** where pH = - log [H⁺] and pOH = -log [OH⁻]
- [H⁺] = 10^{-pH} and [OH⁻] = 10^{-pOH}

14. #1-3 are examples. Complete the rest of the table. No calculator for #1-8!

#	[H ⁺] M	[OH ⁻] M	pH	pOH	pH + pOH	Acid, Base or Neutral
1	1×10^{-12}	1×10^{-2}	12	2	14	base
2	1×10^{-4}	1×10^{-10}	4	10	14	acid
3	1×10^{-7}	1×10^{-7}	7	7	14	neutral
4			5			
5				6		
6						neutral
7	1×10^{-1}					
8		1×10^{-3}				
Now use your calculator and the equations to solve these:						
9			10.5			
10				2.2		
11	2.0×10^{-3}					

U12 Lesson #4 Neutralization and Titration

1. Neutralization reactions are a special form of a _____ reaction.
An Acid and a base react to form an _____ and _____.
2. If a strong acid is reacted with a strong base in equal parts, the resulting solution is _____ (pH = ____).
3. Copy Examples:
acid + base → ionic compound + water
____ + _____ → _____ + _____
acid + base → ionic compound + water
____ + _____ → _____ + _____
4. Titrations are used to determine the _____ of an unknown acid or base.
5. The steps in titration are as follows:
 - a. A measured volume of an acid solution with an unknown concentration is added to a flask.
 - b. Several drops of the _____ are added to the solution.
 - c. Measured volumes of a base of known concentration (called the standard solution) are mixed into the acid using a _____.
 - d. When the solution just barely changes color, the titration has reached its "end point".
 - e. The "endpoint" or _____ point is when the number of moles of hydrogen ions equals the number of moles of hydroxide ions. This is the point when the acid has been neutralized.
6. Example: A 12.5 mL HNO_3 solution is completely neutralized by 25.0 mL of 0.500 M NaOH. What was the concentration of HNO_3 ?

Start by writing the balanced equation:

If (and only if) you have equal moles of OH^{1-} and H^{1+} (a 1:1 acid: base mole ratio) in your formulas, use this equation: $M_{\text{acid}}V_{\text{acid}} = M_{\text{base}}V_{\text{base}}$

Solve:

7. How many milliliters of 0.45 M HCl will neutralize 25.0 mL of 1.00 M KOH?

Balanced Equation:

Can you use $M_{\text{acid}}V_{\text{acid}} = M_{\text{base}}V_{\text{base}}$? Is it a 1:1 ratio?

Solve:

Expert question: How many milliliters of 1.5 M H_2SO_4 will neutralize 20. mL of 1.0 M NaOH?

Hint: Write the balanced equation and then use stoichiometry to solve going through the following steps: (1) find moles of base (2) convert to moles of acid using mole ratio (3) use molarity equation to find the volume of acid

Answer: 6.7 mL